

Battery Health Report - Ikeja Cluster (Quarterly)

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1. Battery System Overview

All three Ikeja towers (TWR-IKJ-007, TWR-IKJ-019, TWR-IKJ-021) are equipped with VRLA (Valve Regulated Lead Acid) battery banks. These banks provide backup power during grid or generator outages and are maintained under standard MTN Nigeria operational procedures.

2. Battery Health Assessment

Tower	Bank Capacity (Ah)	State of Health %	Last Discharge Test	Backup Duration	Condition	Action
TWR-IKJ-007	600	94%	Jan 15, 2026	4.2h	Good	None
TWR-IKJ-019	600	81%	Jan 15, 2026	3.6h	Fair	Monitor
TWR-IKJ-021	600	97%	Jan 15, 2026	4.4h	Excellent	None

3. Discharge Test Results

Controlled load tests were conducted on January 15, 2026. Each battery bank was discharged under a simulated site load, with voltage and current readings taken every 15 minutes. Backup durations were recorded until the cut-off voltage was reached.

4. Cell Voltage Uniformity - TWR-IKJ-019

Cell No.	Voltage (V)
1	2.12
2	2.11
3	2.09
4	2.08
5	2.07
6	2.05
7	2.03

8 | 1.98 (below spec)
9 | 2.02
10 | 1.97 (below spec)
11 | 2.04
12 | 1.96 (below spec)

5. Float Voltage Check

All banks were checked for rectifier float voltage. TWR-IKJ-019 rectifier float voltage was slightly high at 54.8V (specification: 54.0V $\pm$ 0.5V). Adjustment was carried out on site to bring voltage within acceptable range.

6. Electrolyte / Visual Inspection

All batteries were inspected for physical integrity. No swelling, leakage, or corrosion was observed. Terminals were cleaned and tightened. TWR-IKJ-019 showed minor discoloration on three cells, consistent with reduced SOH readings.

7. Replacement Forecast

TWR-IKJ-019 battery bank has an estimated useful life of 18 months remaining. Replacement is planned for Q3 2027 in line with MTN Nigeria's preventive maintenance schedule.

8. Recommendations

- Replace TWR-IKJ-019 cells with SOH <80% (three cells identified).
- Conduct quarterly rectifier float voltage checks to ensure compliance with specification.
- Continue monitoring TWR-IKJ-019 bank closely until replacement schedule.

9. Sign-off

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